

Name of the Student

Reg. No

GitHub Link

D.Chinna Moulali

192 425 078

SIMATS ENGINEERING

CO2-ASSESSMENT TOOL 3

Class Test

COURSE NAME:

Deep Learning

Faculty Dr. Arul Raj A.M

COURSE CODE: MLA04

Slot :

COURSE OUTCOME COVERED

CO 2: Analyze Machine Learning and Deep Learning models by evaluating neural network architectures, representation learning, activation functions, unsupervised learning techniques, and their real-world applications. (L4)

CO Weightage: 15%

Assessment Tool Weightage: 20%

Duration: 90 minutes

Marks: 25

Code of Conduct:

I

10. Chinna Moulali

(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary

1. Differentiate between Machine Learning and Deep Learning with respect to architecture, data requirements, feature extraction and applications. Give two real-world applications of

each.

2. Explain the concept of **Representation Learning**. Discuss how the **width** and **depth** of a neural network influence its learning capability, computational complexity and model performance.

3. Compare **ReLU**, **Leaky ReLU (LReLU)** and **ELU (Exponential Linear Unit)** activation functions. Explain their mathematical behavior, advantages, disadvantages and suitable use

cases.

4. Explain the concept of **unsupervised learning** in neural networks. Describe the working principle of **Restricted Boltzmann Machines (RBMs)** and **Autoencoders** and compare their objectives and applications.

5. Explain the concept of Autoencoder and its uses in Deep learning.

RUBRICS FOR CALCULATION:

Criteria

Weightage

Level 4- Exemplary

Concept

25%

Identifies all key

Identification

concepts from

Linkages

25%

literature

accurately and

comprehensively

Establishes clear,

meaningful, and

accurate

relationships

between concepts

Organization

20%

Concepts are well-structured

Level 3- Proficient

Identifies most key concepts with minor omissions Shows correct relationships with minor gaps

Mostly organized with minor

Level 2- Developing

Identifies limited concepts; some are irrelevant or missing Relationships are weak, unclear, or partially incorrect Some organization but lacks clear hierarchy

Level 1 - Beginning Fails to

identify essential concepts

No

meaningful relationships established

No logical organization or structure

Integration of 20%

Literature

hierarchically with
logical flow and
grouping
Integrates multiple
studies effectively
to show
connections and

structural issues

Shows some

Limited

integration of

integration;

No
integration of

literature

mostly isolated

literature

concepts

evidence

comparisons

Clarity

10%

Concept map is

Generally

Clarity is

Poorly

clear, neat, visually

clear with

reduced;

presented and

structured, and

minor

difficult to

hard to

Class Test

Aspect

Architecture

rise
algorithm

Name: 10 Chinna Moulali

Regno: 1924 2 5 2 7 8

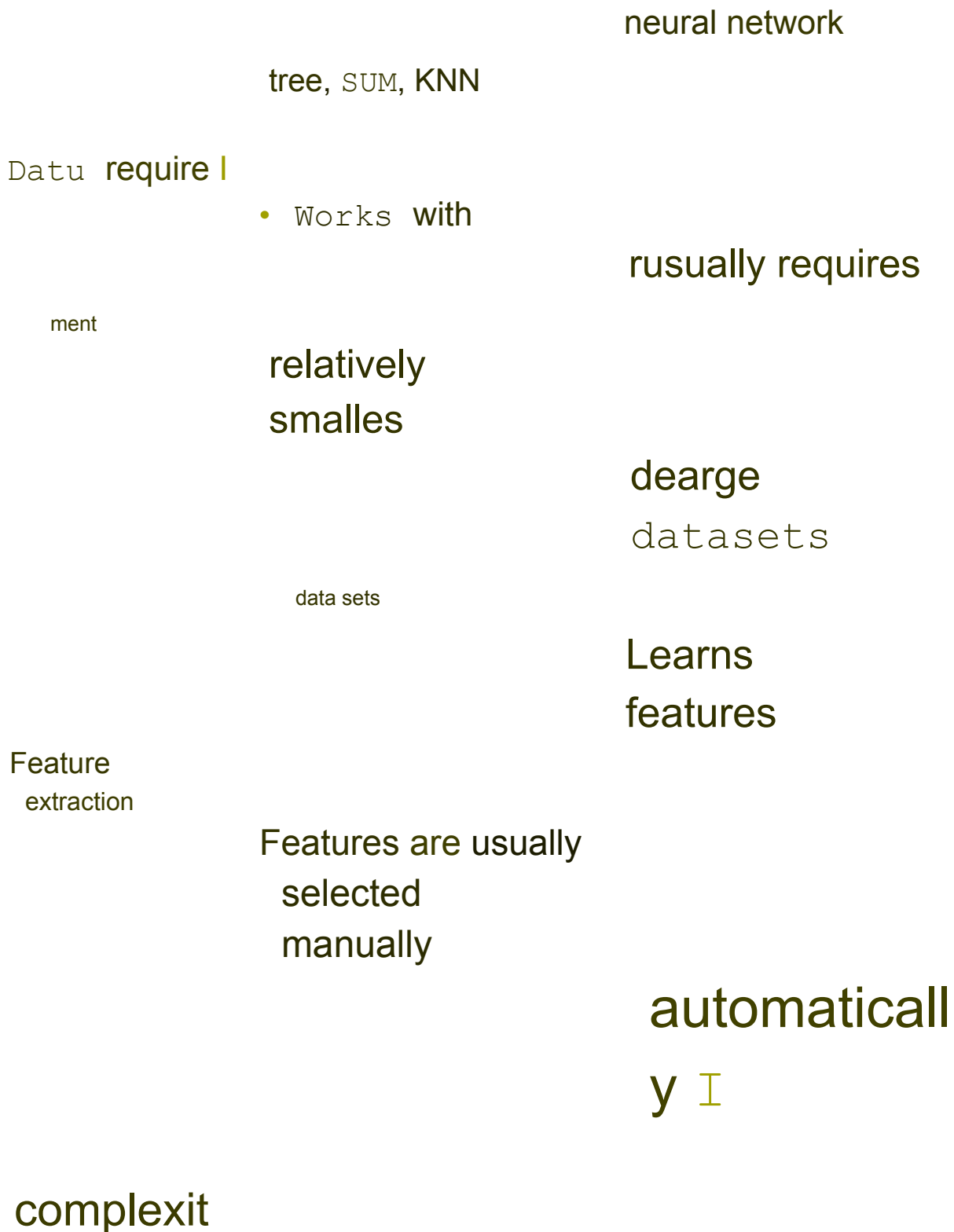
eee

de

Machine learning Deep
learning

such as Decision

uses multi hayer
(DA)



y

complex

computati
on

requirement

comparatively less

High complex

Lowers computational

power often apus

Requires high
computa

Applicati
ons

spam

detection

Coedit

scoring

Images
recognition

speech
recognition

Music

applications

1 credit card fraud
detection,

© Email

Email spam classification...

Deep learning
applications?

1 Face
recognition
self driving cars

ROY

Representation Learning - width and Depth

Representation learning is a

technique where

we learn a model

representation

N

Representation learning is automatically learns useful

features **or** de

froms raw data instead of reguring human to
manulby I

Select

features

Example: In
image becognition

a neural network can leaon

→
pixels →→ edges → shapes
objects → class

Width :

- Width means the
no

gnore neurons

N

ne

woons in a

layer

patterns

→ can learn more

Increases model

capacity

- Too many neurons can increase computation and cause

Overfitting

h: -vept

means the no

- above

layers

overfitting

the no of layers in a neural

network

allows the models to learn hierarchical

and complex
features

Increase

computational complexity

||

overfitting

deep deep learning networks
may suffer from overfitting
if not properly designed

3

Features

ReLU

Leaky

Rho

ELU

Formula

$$f(x) = \begin{cases} x & x \geq 0 \\ \max(0, x) & x < 0 \end{cases}$$

$$f(x) = \begin{cases} x & x \geq 0 \\ 1 & x < 0 \end{cases}$$

main

sad

vantage

Negative

Input

- advantage

simple a

Dying Relu
Requiges a

small

Value

negila
ve

$+(x) = x \cdot 18 \cdot x^{30}$
alex 12 1/x=0
u

smooth

negive

Malue

Reducing
dying Rehu
poblm

smoom

negine

regio

n

Where
computational
• expensive

suitable for

General
deep
network

Network's

Deep
network

Whereby
mg

recurring &
module

ReLU is concern

activation

Reno

ReLU: For

positive

Leaky

ReLU:

positive
value

for

negative

Values; output is zero are less likely to be zero

Allows a small negative
value output,

so

Neurons are less likely

10

he

come

me

peromantly

inactive

Ehoo

Produces
smoom ne

given

en better

negenv output gradient
behavior around
2000.

und

Gene
ally

14 Unsupervized leaoning RBM
and Autoencoder:

It is a type of machine learning where the

model

s patterns form run
labelled data

learns

Escam

ples

les Include;

o

cluster
ing

Dimensionally
reduction

Feature learning 1

Anomaly

detection

Restricted Boltzman machine

(RIBNA) .

1. An, RBAN is a neural network consisting mainly of

Visible layer -

Hidden layer

There are no

connection between

neurons with the same

layer

working:

1. *Input*

data

прис

data is giv

is given to

visble layer

2. Hidden neusons learns

pattern from Input

2:

3. "The model learns the probability distribution

4. It can reconstruct of

Application

S.

-

Recommendation

6

system

S

generate similar data

generate

ate

Feature

learning

Dimensionally
reduction.

data

5

Autoencoder and Its user in
Deep learn

Deep
learnin
g:

An Autoencoder is an unsu
unsupervised neural network that
pro
decons to repacsent input data in a
compressed form

and men deconstruct the

ooigin
al

data

Architecture?

Япри

к

→ Encoder → Latent-space- → Decoder -
Reconstructed

Encoder: converts input to compact representation

latent

output

Space: • stores important features of the input

• Decoder? Reconstructs the

The model

or

original

Input.

Dies to minimize reconstructional

error.

Uses of Autoencoders!

1 Dimensionality reduction ~ reduce
the no

of features

2 Feature extraction - feature important
features au

B) Data.com

automaticall
y

Data compression - stores data using fewer
dimensions

4 Denoising - Removes noise
from image or signals 15 Anomaly

detection - unusual data

©

Image

reconstruction.

generally

has

Reconstructs of

higher

Improves

image

C Representation learning, heaons
useful hidden

Example:

For a noise

-

imag

e

Noisy image → Encoder → latent
space → Decoder → Clean image

feature



latent

representation

learn

/

Clean / Reconstruct
image